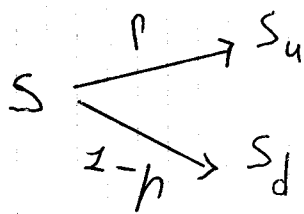


Exercise Lecture 10



$$S = 100$$

$$S_u = 150$$

$$S_d = 75$$

$$p = \frac{1}{2}$$

$$r = 0$$

$$K = 110$$

European call

Real world

$$C_u = \max(S_u - K, 0) = \max(150 - 110, 0) = 40$$

$$C_d = \max(S_d - K, 0) = 0$$

$$C^{\text{Real}} = \frac{1}{1+r} \left(p C_u + (1-p) C_d \right) = \frac{1}{1+0} \left(\frac{1}{2} \cdot 40 + \frac{1}{2} \cdot 0 \right) = 20$$

Risk-neutral

$$q = \frac{S(1+r) - S_d}{S_u - S_d} = \frac{100 - 75}{150 - 75} = \frac{25}{75} = \frac{1}{3}$$

$$C^{\text{Risk-Neutral}} = \left(\frac{1}{3} \cdot 40 + \frac{2}{3} \cdot 0 \right) = \frac{40}{3} \approx 13,33$$

$$\Rightarrow C^{\text{Real}} \neq C^{\text{Risk-Neutral}}$$

Show how you could construct arbitrage strategy if someone would sell you the call at the price computed with real world prob.

$$C^{\text{Real}} = 20 > C^{\text{Risk neutral}}$$

$\Rightarrow$ call is overpriced $\Rightarrow$ sell it and hedge

$$\begin{cases} \Delta^* = \frac{1}{1+2} \left(C_u - \frac{C_u - C_d}{S_u - S_d} S_u \right) \\ \Delta^* = \frac{C_u - C_d}{S_u - S_d} \end{cases}$$

$$\begin{aligned} \Rightarrow \Delta^* &= \frac{40 - 0}{150 - 75} = \frac{40}{75} = \frac{8}{15} \\ \Delta^* &= 1 \left(40 - \frac{8}{15} \cdot 150 \right) = -40 \end{aligned}$$

$$\begin{aligned} \Rightarrow H^* &= \Delta^* + \Delta^* S = -40 + \frac{8}{15} \cdot 100 \\ &= 13,33 \\ &= C^{\text{Risk neutral}} \end{aligned}$$

$t=0$:

Sell at C^{Real}

Buy Δ^* shares at 100

Borrow $\Delta^* = 40$

CF's

+20

-53,33

+40

+6,77

$t = 1$
call
shares

up ($S = 150$)
 $110 - 150 = -40$
 $\frac{8}{15} \cdot 150 = 80$

down ($S = 75$)
 0 ($75 < 110$)
 $\frac{8}{15} \cdot 75 = 40$

Repay loan

+ $\frac{-40}{0} \quad \frac{-40}{0}$

$\Rightarrow$ The replicating portfolio cancels your obligation in every scenario and you locked in 0,67 at $t = 0$